
Lab 3: Adaptive LUT

Goals

- Understand the use of multiplexers as **LUTs**.
- Explore different methods for designing digital circuits.

Introduction

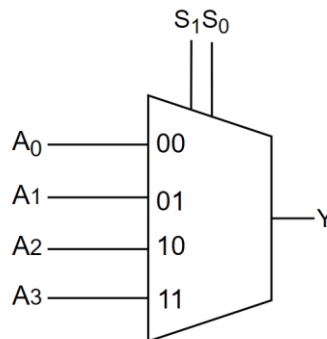
In this lab, an **Adaptive Look-Up Table (LUT)** will be implemented.

A **LUT** is a storage structure that allows mapping inputs to predefined outputs. In digital systems, LUTs are used to implement logic functions, perform data transformations, or store values for quick access. An **adaptive LUT** enables the values to be updated during runtime, increasing flexibility for applications such as:

- Implementation of **custom mathematical functions**.
- Algorithms that require **variable configurations**.
- Systems with **real-time reconfiguration**.

Lab description

A **LUT** can be implemented using multiplexers. The **multiplexer inputs** can be interpreted as the predefined entries of the LUT, and the **selector** chooses one of these predefined entries and propagates it to the output.



The following **truth table** illustrates the behavior of a **4-input multiplexer**:

S1	S0	Y
0	0	A0
0	1	A1
1	0	A2
1	1	A3

From the **truth table**, we observe that **A** determines the **LUT output**, thus defining the circuit's behavior.

If the values of **A** can be modified, then we can **alter the circuit's behavior** dynamically. A circuit that allows LUT values to be updated is known as an **adaptive LUT**.

Step 1.

Design an **adaptive LUT** capable of selecting between **8 values**.

Step 2.

Implement the circuit on the **FPGA**.

- Use **switches SW[0] to SW[7]** for input data.
- Use **switches SW[15] to SW[17]** as the selector.
- Use **LED LEDR[0]** for the output bit.

Step 3

Implement the following **truth tables**.

LUT 1

S2	S1	S0	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

LUT 2

S2	S1	S0	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Submission

Implementation description

Generate a document providing a detailed description of the implementation. This should include an explanation of design decisions, challenges encountered, and how they were overcome. Include diagrams, schematics, tables, or equations used in the design development.

Video:

Upload a short video to the Teams group, recording the operation of the implemented circuit on the FPGA and providing a brief description of its functionality.
